

STM32L0 MODULAR DEVELOPMENT & PROTOTYPING PLATFORM

Full Hardware Project Documentation

Main MCU Board + DB37 Connection / Breakout Board + Firmware-ready Platform

A reusable STM32L0 hardware platform designed for firmware development, peripheral experiments, sensor integration, interface testing, and rapid prototyping.

Primary MCU	STM32L062K8T6 (LQFP32)
Platform concept	Reusable STM32L0 development and prototyping system
Hardware status	Schematic completed, PCB completed, board assembled and used
Expansion	37-pin board-to-board connection to a dedicated breakout / connection PCB
Main interfaces	SWD, USB, UART, SPI, I2C, GPIO, ADC, DAC, EXTI
Documentation scope	Main board schematic, connection board schematic, pinout, and jumper configuration

Project positioning

This is a full hardware development project rather than a small demonstration board. The platform combines the MCU core, power circuitry, analog conditioning, communication interfaces, configurable routing, a dedicated connection board, and supporting documentation.

1. Project Purpose

The platform was created to avoid rebuilding the same MCU support circuitry for every prototype. Instead, the main board provides a reusable STM32L0 core with common interfaces and analog resources already available, while a separate DB37 connection board makes the signals easy to access during laboratory work and firmware development.

The assembled implementation uses the STM32L062K8T6. The design concept can also be used with other STM32L0 devices that are pin-compatible with the selected LQFP32 footprint. Exact device substitution should be checked against the specific MCU datasheet and alternate-function mapping before use.

System Concept

MAIN MCU BOARD	DB37 INTERCONNECT	CONNECTION BOARD
STM32L0 core Power regulation Analog conditioning USB / SWD / UART / SPI / I2C Jumpers and local peripherals	37-pin link Carries MCU and peripheral signals Keeps the main board reusable	Breaks signals into accessible groups GPIO / ADC / DAC UART / SPI / I2C SWD / USB Power rails and test connections

2. Main Hardware Functions

MCU core: STM32L062K8T6 with GPIO and alternate-function access routed for prototyping.

Programming and debug: Dedicated ST-LINK/SWD connection using SWCLK, SWDIO, 3.3 V and GND.

Digital communication: USART, SPI and I2C signals available for external modules and firmware experiments.

USB: USB D+ and D- lines are available on the platform.

Analog section: Multiple ADC paths, DAC access and operational-amplifier stages based on TLV2374/TLV2372 devices.

I/O expansion: PCA9654 I2C GPIO expander provides additional digital I/O resources.

Sensors and modules: Connections include a motion sensor input, IR receiver and a JDY communication-module interface.

Clocking and system support: 32.768 kHz crystal provision, reset circuitry, BOOT0 control and configurable jumpers.

Power: 12 V input with on-board 3.3 V regulation and local decoupling for the MCU and peripherals.

3. Prototyping and Configuration Strategy

The platform is designed around configurable routing rather than a single fixed application. Jumpers and 0-ohm links allow selected MCU pins to be connected to analog channels, sensor functions, or external connectors as required. This makes the same assembled board useful across different firmware and hardware experiments.

Area	Examples	Purpose
Analog routing	PA0, PA1, PA4, PA5, PA6 / ADC and DAC paths	Select direct GPIO or analog-function routing.
Sensor routing	PA7 / motion input	Reuse the same MCU pin for different test configurations.
I2C	SCL1 / SDA1 and optional pull-up connection	Support external I2C devices while avoiding duplicate pull-ups when necessary.
Expansion I/O	P4-P7 from the I2C expander	Add accessible digital I/O without consuming additional MCU pins.
Board-to-board access	DB37 plus grouped external connectors	Bring MCU and peripheral signals to the connection board for practical laboratory access.

4. Firmware and Real-Project Use

The hardware platform has already been used in a real engineering application, demonstrating that the design moved beyond schematic and PCB development into practical use. The original employer-related firmware is not included in this public project documentation.

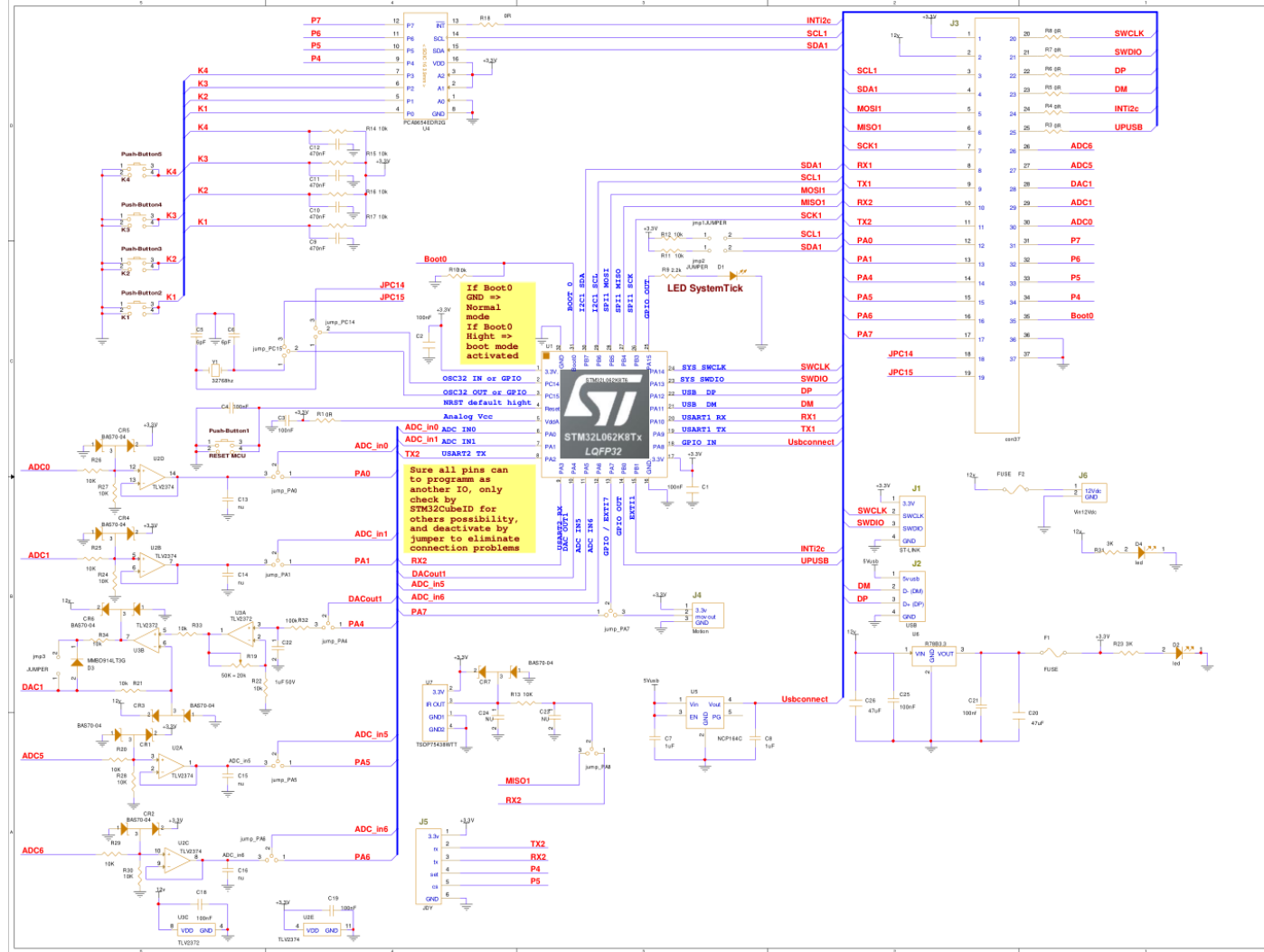
For a public GitHub repository, an independently developed firmware example can be provided instead. A suitable demonstration can exercise the STM32L062 peripherals and the board resources, for example GPIO, ADC, DAC, UART, I2C, SPI, buttons, interrupts and the motion-sensor input.

Public repository note

The public repository should contain only hardware files and firmware that are owned and safe to publish. Proprietary application code should remain excluded.

Appendix A - Main STM32L0 Board Schematic

Main controller board centered on STM32L062K8T6 with power, analog, communication, programming and expansion circuitry.

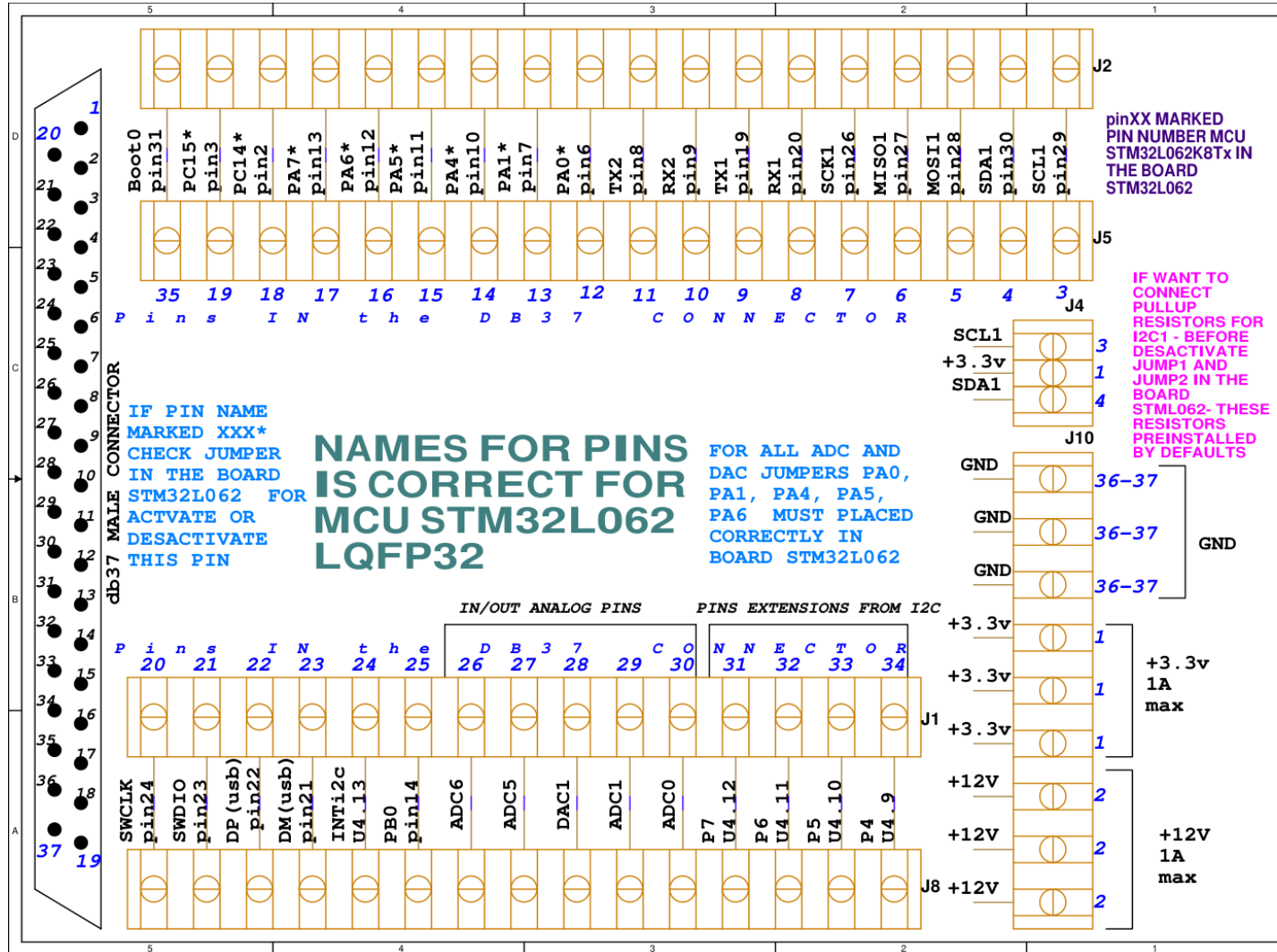


Main-board schematic. The large 37-pin connector provides the link to the separate connection / breakout board.

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Appendix C - Connection Board Pinout

Pin mapping for STM32L062 signals, analog functions, communication interfaces, power rails and I2C-expander outputs.



Appendix D - Jumper and Button Configuration

Configuration reference for jumpers, analog routing, motion input and local push buttons.

	5	4	3	2	1
1					
2	SWITCH	PIN 1	PIN 2	PIN3	INFO
3	JUMP1	X	X	N/A	on/off RESISTOR 10K SCL1
4					
5	JUMP2	X	X	N/A	on/off RESISTOR 10K SDA1
6					
7	JUMP3	X	X	N/A	on/off DIODE DAC1
8					
9	JUMP PA0	1-2 PA0 ready	2-2 off ALL	3-2 ADC0 ready	CONNECT PA0 or ADC0 TO EXT
10					
11	JUMP PA1	1-2 PA1 ready	2-2 off ALL	3-2 ADC1 ready	CONNECT PA1 or ADC1 TO EXT
12					
13	JUMP PA4	1-2 PA4 ready	2-2 off ALL	3-2 DAC1 ready	CONNECT PA4 or DAC1 TO EXT
14					
15	JUMP PA5	1-2 PA5 ready	2-2 off ALL	3-2 ADC5 ready	CONNECT PA5 or ADC5 TO EXT
16					
17	JUMP PA6	1-2 PA6 ready	2-2 off ALL	3-2 ADC6 ready	CONNECT PA6 or ADC6 TO EXT
18					
19	JUMP PA7	1-2 PA7 ready	2-2 off ALL	3-2 MOTION ready	CONNECT PA7 or MOTION READY
20					
21	JUMP PA8	1-2 RX2(PA3) ready IR	2-2 off ALL	3-2 MISO1(PB4) ready IR	CONNECT PA7 or MOTION READY
22					
23	Push Button1				reset mcu restart
24					
25	K1				press P0=0 U4(i2c) default = 1
26					
27	K2				press P1=0 U4(i2c) default = 1
28					
29	K3				press P2=0 U4(i2c) default = 1
30					
31	K4				press P3=0 U4(i2c) default = 1
32					

Jumper / switch configuration table used when adapting the platform to different prototype setups.

6. Project Status and Next Steps

The hardware platform is already designed, manufactured and assembled. The next public-development step is to package the hardware documentation together with an independent firmware demonstration and photographs of the assembled boards.

- Add photographs of the assembled main board and connection board.
- Add a short bring-up procedure: power, SWD connection, COM / UART test and initial firmware load.
- Publish independent demonstration firmware for the STM32L062.
- Document one or more practical examples, such as the motion-sensor input, ADC acquisition or an I2C peripheral test.
- Keep the main platform reusable so that future examples can be added without redesigning the core hardware.

Project classification: Full Hardware Project - STM32L0 Modular Development & Prototyping Platform

Prepared as project documentation for GitHub and technical portfolio use.